CONCEPT OF UNCONSCIOUS-DEFORMATION ASSESSMENT FOR FORMATIVE E-ASSESSMENT DESIGNERS

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Abstract

This work-in-progress paper discusses an issue of how to assess unconscious-deformations. The issue is emerged in Formative e-Assessment (*FA*) design workshops to approach a phenomenon of parallel low-literacies that face German educators and STEM adult-migrant educatees migrated to Germany for study, work, or asylum purposes. The paper describes three parallel low-literacies: namely algorithms low-literacy, German-language low-literacy, and self-assessment low-literacy. The paper reports on how the workshops are organized to enable building an adult-education society with and for adult-migrants suffered from low-literacies and submerged in a culture of silence. The paper presents an innovative concept of Unconscious-Deformation (*UD*) assessment. It presents how the concept's main functionality is designed to enable explorative-questioning driven assessment of unconscious-deformations. The paper describes the two core components of *UD* assessment concept: namely, theoretical and operational components. The theoretical-component is grounded on integrating Miller's pyramid of assessment and Paul Freire's theoretical-notions "literacy as power, culture of silence and critical-consciousness". Freire's theoretical-notions are for approaching the adult-educatees' culture of silence as symptom of invisible oppressions that need to be analyzed. The operational-component is designed as easy deployable tool to enable explorative-questioning driven assessment. The tool is to serve online-capturing and visualizing of adult-educatees' explorative questions in a host-guest setting for multi-level cultural-translations. The setting helps *FA* designers simulate the multi-level cultural-translations between guest-expressions and host-expressions. The host-expressions are expressions of a German educator, whereas guest-expressions are expressions of an adult-migrant educatee. Moreover, the paper describes how explorative questions are captured to support researching Multi-Level Cultural-Autoencoder (*MLCA*) algorithms. The *MLCA* algorithms are neural-net algorithms suggested to help the *FA* designers in investigating how an adult-migrant educatee can codify, recodify and decodify his/her unconscious-deformations in descriptive and comparative activities for mastering his/her self-assessment. To test the design of the suggested *UD* assessment concept, test cases are suggested as next step of this work-in-progress paper. Finally, for reflection and feedback exchange with the ICERI community we summarize the limits of the suggested concept and the future work to improve it.

Keywords: Formative E-Assessment, Unconscious-Deformation, Miller's Pyramid of Assessment, Autoencoder Algorithms, Literacy as Power, Culture of Silence.

1 INTRODUCTION

Assessment is usually understood to have two purposes: summative and formative. Summative assessment is used to judge students' achievements at the end of an instructional unit, whereas formative assessment aims to gather evidence about students' learning progress to influence and guide teacher's approaches, pedagogies, and priorities. To support educators, teachers, and trainers interested in formative assessment to serve STEM adult-migrants migrated to Germany for study, work, or asylum purposes, this paper introduces our workshops-based practice for designing Formative e-Assessment (***FA***) as inquiry. The main goal of our practice is to help the *FA* designers (i.e., trainee-teachers, teachers, trainers or adult-migrants as self-teachers) understand better the issues of parallel low literacies that face the adult-migrants. As requirement to present our practice we present first the literature review.

2 LITERATURE REVIEW

According to Hattie[1], using two notions (formative vs summative) is not helpful in educational debate about assessment and visible learning [1]. To improve the debate Hattie suggests using the single notion

[1] John Allan Clinton Hattie (born 1950) is a New Zealand education academic. He has been a professor of education and director of the Melbourne Education Research Institute at the University of Melbourne, Australia, since March 2011.

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"Assessment-capable learner". This notion enables him to argue that both forms of assessments: summative and formative are important. To support his argument, he used the metaphor suggested by Robert Stake[2] that draws not only a clear distinction line between formative and summative assessments but also highlights the importance of them both. The metaphor is: "*When the cook tastes the soup, that's formative and when the guests taste the soup, that's summative*". In their book "Turning Point for the Teaching Profession" [2] Hattie, Rickards, et al are arguing for the establishment of teaching as a true "profession" alongside areas such as medicine or law. Moreover, in their book, they are elaborating on evaluative thinking and clinical practice as the basis of this new profession. Hattie's Visible Learning [1] pedagogy was developed in context of primary schooling, therefore in our research we aim to test if his pedagogy and its explaining constructs hold or do not hold in contexts of our offers for groups of freshman students or/and migrant students in a first year STEM university program.

In our previous works [3, 4, 5, 6, 7, 8] we research **F**ormative e**A**ssessment (***FA***) tasks in STEM university contexts. Our research is about how to understand the pedagogy-assessment linkage and how to approach the research based *FA* challenging-tasks contextualized in different STEM teaching or training settings. Examples of these tasks are: a task contextualized in a computer organization course that has about 130 freshman students [3], a task contextualized in training projects for big-data analytics [4], and a task contextualized in workshops for developing deep-learning APIs [5]. In contrast to our previous works in [3, 4, 5, 6, 7, 8] that do not focus on the issues of parallel low literacies that face the adult-migrants in Germany, these issues are the focus of this paper.

To approach and understand literacy, the literacy researcher Brian Street[3] suggested to distinguish between two approaches: autonomous approach and ideological approach [9,10]. The autonomous approach is grounded on an assumption that literacy, autonomously, will have beneficial effects beyond particular literacy events. In Street's words [9], the autonomous approach anticipates that "introducing literacy to poor, 'illiterate' people in villages, urban youth etc. will have the effect of enhancing their cognitive skills, improving their economic prospects, making them better citizens, regardless of the social and economic conditions that accounted for their 'illiteracy' in the first place".

The ideological approach, on the other hand, conceives literacy as inseparable from the context in which it is enacted. Literacy, according to this approach, is seen as embedded in social practices with their respective socially and historically developed epistemological traits. To be literate in one sphere of life is thus, according to the ideological approach, different from being literate in another sphere, which, by the way, explains why advocates of this approach speak in terms of literacies, in the plural. Literacy practices and meanings of literacy are linked to the issue of what constitutes worth knowing in a certain social practice and how knowledge and identity are performed by the people in the community of practice where literacy is enacted [11].

Freire initiated a critical approach to literacy [12], focusing on adult education and promoting the notion that "reading the world precedes reading the word". In the book Literacy: Reading the Word and the World, written with his long-time collaborator Donaldo Macedo, Freire explains that the organization of literacy programs should be based on what he calls the "word universe" of learners. By reading the students' world and identifying the most meaningful words in their existential experiences, educators come to recognize how they express their actual language, anxieties, fears, demands, and dreams. The learning process happens through "codifications" i.e., pictures representing situations related to learners' daily lives. Reading, or "decoding" the images of concrete situations enables learners to reflect on their former interpretation of the world before going on to read words. More than becoming literate, they learn to read situations of injustice to become agents of change; they insert themselves in the world as capable of subverting a reality that is not favorable to them [13].

By redefining and politicizing the notion of literacy [14, 15, 16], Freire developed a type of critical analysis in which he asserted that traditional forms of education functioned primarily to reify and alienate oppressed groups: Moreover, Freire in "pedagogy of the oppressed" [14] explored in great depth the reproductive nature of dominant culture and systematically analyzed how it functioned through specific social practices and texts to produce and maintain a '"culture of silence" among Brazilian peasants. Though Freire did not use the term "*hidden curriculum*"[4] as part of his discourse, he demonstrated

[2] Robert E. Stake (born December 18, 1927) is a Professor Emeritus of Education at the University of Illinois, Urbana-Champaign. https://en.wikipedia.org/wiki/Robert_E._Stake, retrieved on 10.07.2024.

[3] Brian Vincent Street (24 October 1943 – 21 June 2017) was a professor of language education at King's College London and visiting professor at the Graduate School of Education in University of Pennsylvania. During his career, he mainly worked on literacy in both theoretical and applied perspectives and is perhaps best known for his book Literacy in Theory and Practice (1984).

[4] John Dewey [21] (1859-1952) explored the hidden curriculum of education in his early 20th century works, especially in his classic, Democracy and Education.

pedagogical approaches through which groups of learners could decide ideological and material practices, and in the form, content, and selective omissions of these (i.e., practices) one uncovered the logic of domination and oppression [14, 16].

How to look at and analyze the logic of oppression in pre-migration or/and migration context that hinders the adult-migrant learners from building their self-assessment and self-formation is one of the focus points in this research paper. This paper focuses mainly on a research question emerged in context of a workshop setting for understanding the parallel low-literacies that face the adult-migrant leaners migrated to Germany from Middle east. The workshops setting and its emerged research question are presented in next section.

3 WORKSHOP SETTING AND ITS EMERGED RESEARCH QUESTION

Fig. 1 presented a diagram to enable clarifying the workshops setting in which the research question has emerged. The setting shows two target groups of adult-learners who are interested in reflecting on the barriers they face as *FA* system users.

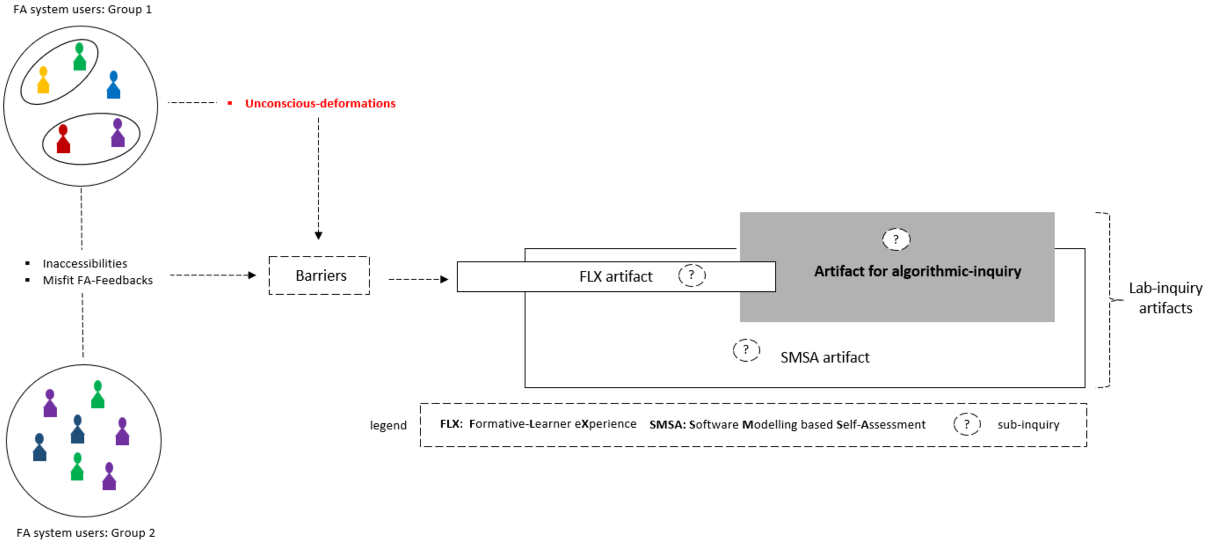

Fig 1. Workshops setting: Lab inquiry artifacts for discussing the barriers of FA system users.

The shared barriers observed in context of the target groups 1 and 2 in Fig. 1 are barriers of inaccessibility and misfit *FA*-feedback. Besides the sharded barriers, the barriers of **U**nconscious **D**eformation (***UD***) are observed only in context of the target-group 1. The target group 1 is the focus of this paper. It is a group of adult-migrant learners in Germany interested on understanding and overcoming the barriers they face in their pre-migration or migration context.

The workshops setting in Fig .1 enables introducing the adult-migrant learners to our research in redesigning the *FA* systems. Briefly it enables introducing them how the research is directed to understand the issues of parallel three low-literacies**:** namely algorithms low-literacy, German-language low-literacy, and self-assessment low-literacy**.** The issues are observed as barriers in a list of other barriers that face the adult-migrant learners due to the oppressive or repressive conditions in their pre-migration or migration context. Their context might have also other complex barriers that hinder them from building their self-assessment and self-formation for their future careers.

In scope of their paper the *UD* barriers are the focus. To understand the *UD* barriers, a research question has emerged. The question is:

- Can the *UD* barriers be approached as objects of formative assessment? If yes, how to develop a concept of *UD* assessment in pre-migration or migration context of the target-group-1?

To enable clarifying the "*UD* barriers" term in the question formulation, the workshops setting in Fig. 1, shows the barriers and how they are represented in relation to the lab-inquiry artifacts. Fig. 1 shows also how the barriers are initially collected and structured by an artifact for studying the **F**ormative **L**earner e**X**perience (***FLX***). Technically seen, the *FLX* artifact is a **U**ser e**X**perience (***UX***) artifact designed for discussing with the adult-migrant learners the barriers they have experienced. The *FLX* artifact enables discussing these barriers in context of discussing the issue of German-language low-literacy.

To enable discussing the algorithms low-literacy issue, an artifact for algorithmic-inquiry is initially designed and integrated with the *FLX* artifact as seen in Fig. 1.

Moreover, to enable discussing the issue of self-assessment low-literacy, an artifact for **S**oftware **M**odelling based **S**elf-**A**ssessment (***SMSA***) is initially designed and integrated as seen in Fig. 1.

The three artifacts for *FLX*, *SMSA* and algorithmic-inquiry enable formulating the three sub-inquiries of the lab-inquiry. The three sub-inquiries in brief are:

1 Algorithmic inquiry: This inquiry helps the *FA* designers design algorithms in context of a pre-migration or migration context. It has a high priority in scope of this paper because the algorithms low-literacy issue is the first issue we aim to approach.

2 *FLX* inquiry: This inquiry is a type of *UX* inquiry to understand the adult-migrant learners' interaction experiences with a *FA* system they have used in their pre-migration or migration context. The interaction experiences under study are limited to natural- language interactions. This inquiry has the second priority in scope of this paper because the German language low-literacy issue is the second issue we aim to approach.

3 *SMSA* inquiry: This inquiry enables approaching the self-assessment low-literacy issue in an adult-migrant learner context. The inquiry has the third priority in scope of this paper. It uses techniques of software modelling to stimulate an adult-migrant learner interested in critically perceiving and analyzing his/her self-assessment tasks**.**

To put it differently, the lab-inquiry and its three sub-inquires are to help the *FA* designers (e.g., trainee-teachers, teachers, trainers, or adult-learners as self-teachers/self-trainers) redesign their *FA* systems to approach the issues of the three parallel low-literacies.

Finally, clarifying the lab-inquiry, its sub-inquiries, and their artifacts in context of the target-group 1, as seen Fig. 1, is essential for organizing our workshops. The workshops are organized reiteratively. Few of them are organized for special individual needs of an adult-migrant learner and most of them are organized for a sub-group of the target-group 1. The organization of workshops aims to enable building an adult-education society with and for the adult-migrants who suffered from parallel low-literacies and submerged themselves in a culture of silence. To serve the adult-education society, an inquiry methodology is required. The methodology is presented in next section.

4 METHODOLOGY

Fig. 2 shows in a big picture the research methodology. The methodology is to enable lab-inquiry modelling for a *FA* designer role. The *FA* designer role is conceptualized to serve a teaching or self-teaching setting for adults (concrete example is the workshops setting presented in section 3). The setting uses a preexisting *FA* system that needed to be redesigned for approaching a phenomenon of parallel low-literacies that face the adult-migrant learners in Germany.

As seen in Fig. 2, the methodology is based on two components, namely

- The theoretical foundation component. It enables the required interdisciplinary theoretical investigation. The investigation is limited to two disciplinaries, namely:
    - Software Development (***SD***) research disciplinary: Its inquiries in scope of this paper are limited to researching the **M**ulti-**L**evel **C**ultural-**A**utoencoder (*MLCA*) algorithms. The need for researching these algorithms emerged as highly required investigation to serve the suggested ***UD*** assessment concept (see section 5 for more details)
    - Critical pedagogy disciplinary: Its inquiries in scope of this paper are limited to researching how to integrate Miller's model of assessment and Paul Freire's theoretical models to approach oppressions in a literacy setting for adult-migrants. (see section 5.1 for more details).
        - The lab-inquiry component. It has three sub-inquiries for approaching the three issues of the parallel low-literacies observed in our workshops setting presented in section 3.

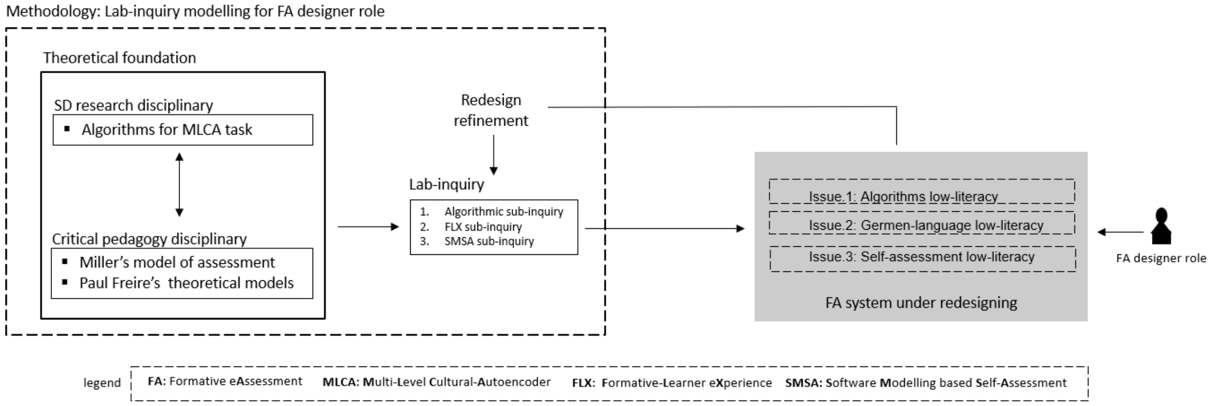

Fig 2. Methodology in big picture.

The role of an *FA* designer in Fig. 2 clarifies how a *FA* system under reengineering can be served by the research methodology. As seen in Fig. 2 the methodology enables lab-inquiry based redesigning process. This process is reiterative and via its iterations it enables redesign refinement as a feedback-loop for lab-inquiry improvement. The lab-inquiry improvement is a goal defined to serve inquiry-based construction of the *UD* assessment concept. The concept is presented in next section.

5 CONCEPT OF UNCONSCIOUS-DEFORMATION ASSESSMENT

Figure 3.1 enables clarifying the suggested concept of Unconscious Deformation (*UD*) assessment and its *UD* diagnosability requirements in big picture.

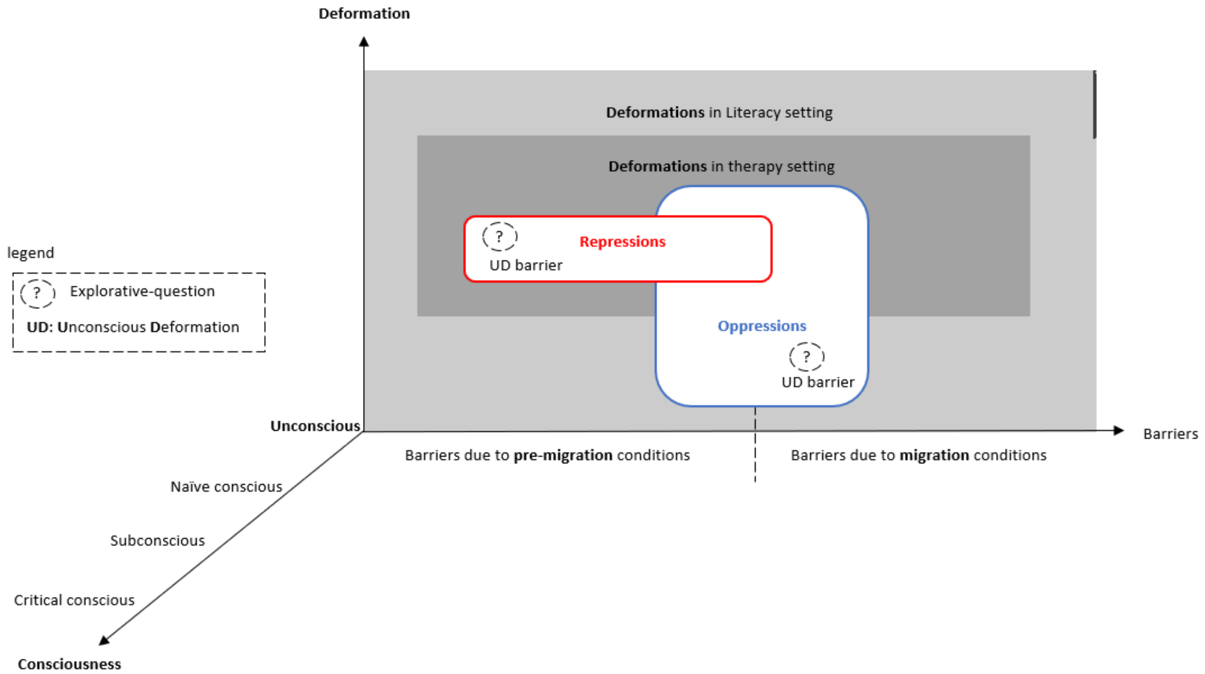

Fig 3.1 how to diagnose UD barriers in settings for literacy or/and therapy.

The figure shows how the concept's main functionality is designed to enable explorative-questioning driven diagnosing of *UD* barriers. The *UD* barriers should be approached and diagnosed as phenomenon of repressions in the therapy setting**,** whereas in the literacy setting they are approached and diagnosed as phenomenon of oppressions.

Moreover, the pre-migration conditions or/and migration conditions behind the *UD* barriers should be also analyzed and diagnosed. The consciousness dimension in figure 3.1 enables studying the deformation phenomena that might be unconscious, naive conscious, subconscious, or critical conscious in relation to the adult-migrant learners' perception. This paper is limited to study the unconscious deformation phenomena. To clarify the *UD* diagnosability requirements, Figure 3.2 shows how the requirements are divided based on the three different functionalities for the three specialist sub-roles of the *FA* designer role.The three roles in brief are:

- Therapy specialist role: This role requires designing for *UD* diagnosability to serve therapy goals at individual level. The goals are driven by the **R**esearch **Q**uestion **1** (**RQ-1**) formulated as follows:
  - How to understand and diagnose the repressions that hinder an individual adult-migrant learner from building his/her self-assessment and self-formation?
- Literacy specialist role: This role requires designing for *UD* diagnosability to serve special literacy required for a group of oppressed adult-migrants. The social literacy goals are driven by the **R**esearch **Q**uestion **2** (**RQ-2**) formulated as follows:
  - How to understand and diagnose the oppressions or/and repressions that hinder a group of oppressed adult-migrants from building their self-assessment and self-formation?
- Compositionist role: this role is a cross-role as seen in Figure 3.2. It is driven by researching how to formulate explorative-questions to serve RQ-1 and RQ-2.

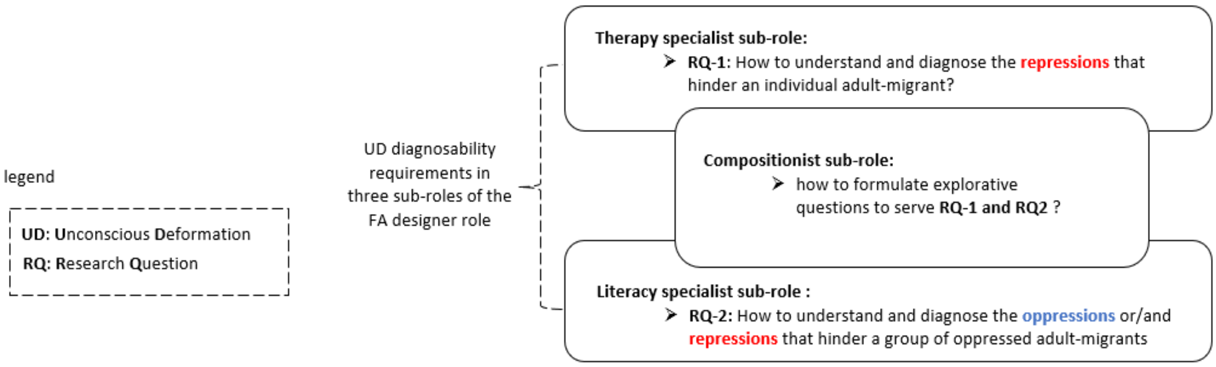

Fig 3.2:UD diagnosability requirements: three specialist sub-roles of FA designer role

The *UD* diagnosability requirements of the three special sub-roles serve a special diagnostic competence for teaching **T**herapy, STEM **L**iteracy and **C**omposition (***TLC***). To put it differently, the *TLC* diagnostic-competence is a special teaching competence that differentiates our research workshops-setting for theory-based FA practicing. The setting has theoretical and operational components presented in next subsections.

5.1 Theoretical Component

Fig. 4 shows, in big picture, the theoretical-component to serve the *UD* assessment concept under inquiry-based construction. This big picture presents the component as theoretical service for *FA* designers. It shows the theoretical notion "*UD* barriers" as object-of-assessment. It also shows how a few of Miller's pyramid based theoretical constructs and a few of Freire's theoretical constructs are selected and integrated to enable finding and synthesizing suitable theoretical constructs to serve the inquiry-based construction of the *UD* assessment concept. The object-of-assessment "*UD* barriers" is mediated between Millier's theoretical constructs and Freire's theoretical constructs. This enables the *FA* designers (e.g., trainee-teachers) to look at teachings that have "*UD* barriers" from two perspectives: a perspective of teaching as clinical practice and a perspective of teaching as political practice. The political practice operates on a collective-level whereas the clinical practice operates on an individual-level.

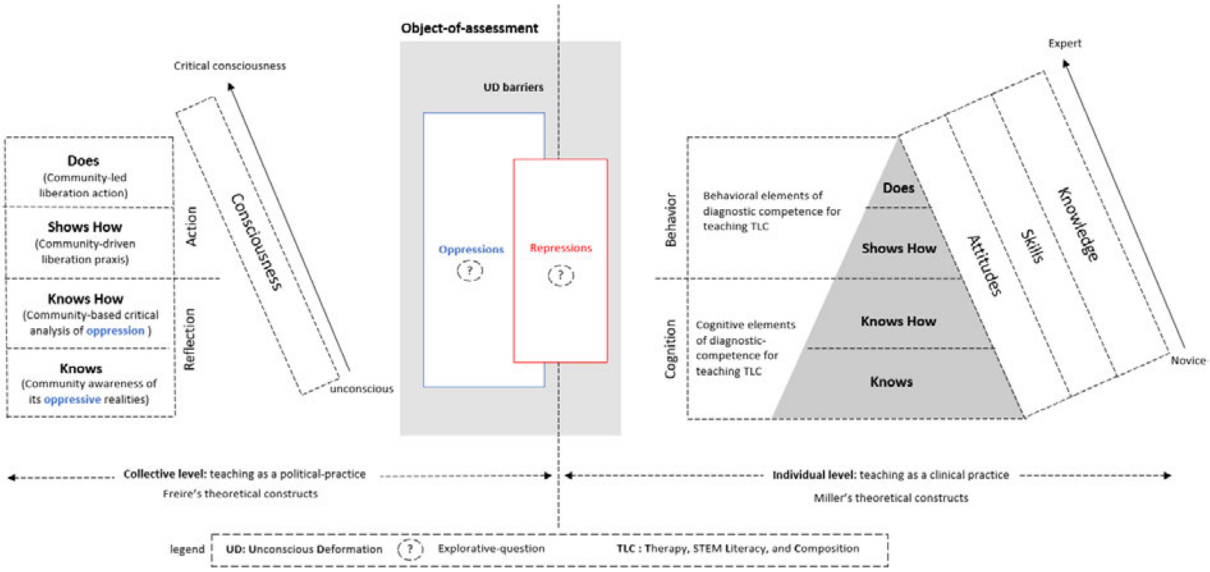

Fig. 4: Theoretical-component as service for FA designers: "UD barriers" as object-of-assessment

The component provides also theoretical-notions for "oppressions" and "repressions" as instances or forms of the object-of-assessment "*UD* barriers". The ""repressions" object is observed at individual-level or/and collective-level, whereas the "oppressions" object is observed at collective-level. To study and analyze the root-cause of "oppressions" and "repressions" at collective-level, the theoretical-component integrates few of Freier's theoretical constructs whereas to study and analyze the root-causes of the "repressions" at individual-level, the theoretical-component integrates few of miller's pyramid based theoretical constructs for assessment as seen in Fig. 4.

Miller's assessment pyramid is designed for assessing the "clinical competence in medical education". In Fig. 4, the notion "clinical competence in medical education" is replaced by the notion "*TLC* diagnostic competence". This notion is also a core notion of the theoretical-component. It refers to the diagnostic-competence required for teaching **T**herapy, **L**iteracy and **C**omposition (*TLC*). In miller's terms, it has cognitive and behavioral elements that should be assessed at individual-level. At collective-level, using Freire's terms, it has elements of reflection and action by a group of oppressed adult-learners. Via their action-reflection-action reiterative work, the oppressed should explore, critically analyze and overcome the oppressive realities that hinder their liberation.

Fig. 4 shows also a guideline to enable the *FA* designers map between Miller's theoretical notions and Freier's theoretical notions. For example, see the mapping for the notions "knows", "knows how", "shows how", "does" as clarified in Fig. 4. Fig. 4 clarifies also the mapping of Miller's notions "cognition and behavior" to Freier's notions "reflection and action".

Not only guidelines for mapping the notions, but also guidelines to construct the dimensions of the object-of-assessment "*UD* barriers" can be provided by the theoretical component. In Fig. 4 there are three dimensions "skills", "attitudes" and "knowledge" that serve designing assessments at individual level using Miller's theoretical constructs. Fig. 4 has also one dimension "consciousness" that serves designing assessments at collective-level using Freier's theoretical constructs.

Designing a theoretical-inquiry that uses Freier's notion "culture of silence" as symptom of invisible oppressions that need to be analyzed is a concrete example for how a *FA* designer can initialize a lab-inquiry based *FA* practice for understanding a group of silent or oppressed adult-migrants. To improve his/her theoretical-inquiry, the *FA* designer can also use Freier's notion "literacy as power" as designer's analytical notion to analyze and explain the low-literacies that face a silent or oppressed group.

To sum up, this section has summarized in big-picture how the *FA* designers can use the theoretical component as service for designing their theoretical-inquires. The section summarizes the theoretical component not only in clarification texts but also in a visual diagram as seen in Fig. 4. Due to the limits of space for this section, many visual diagrams for clarifying the theoretical component's services are not presented. These diagrams can be presented as digital material to help *FA* designers interested in designing lab-inquiry based *FA* practices. Designing the practices requires also operational services provided by the operational-component presented in next section.

5.2 Operational Component

Fig. 5 shows, in big picture, how the operational-component is designed as web tool to serve online-capturing and visualizing *FA* explorative questions in a host-guest setting. The setting is an improved setting of the workshops setting presented in *section 3*. It helps *FA* designers simulate the multi-level cultural-translations between guest-expressions and host-expressions. The host-expressions are expressions of an educator, teacher or trainer in Germany, whereas guest-expressions are expressions of an adult-migrant learner.

Fig. 5 shows how the tool is used to deploy two artifacts; artifact for cultural-translation based *FLX* inquiry and artifact for algorithmic-inquiry. The algorithmic-inquiry is configured to investigate how the explorative questions can be captured to support researching *MLCA* algorithms. The *MLCA* algorithms are neural-net algorithms suggested to help the *FA* designers in investigating how an adult-migrant learner can codify, recodify and decodify streams of his/her repressive or oppressive cultural-expressions. The streams are analyzed as streams of *UD* barriers that hinder the adult-migrant learners from building their self-assessment and self-formation.

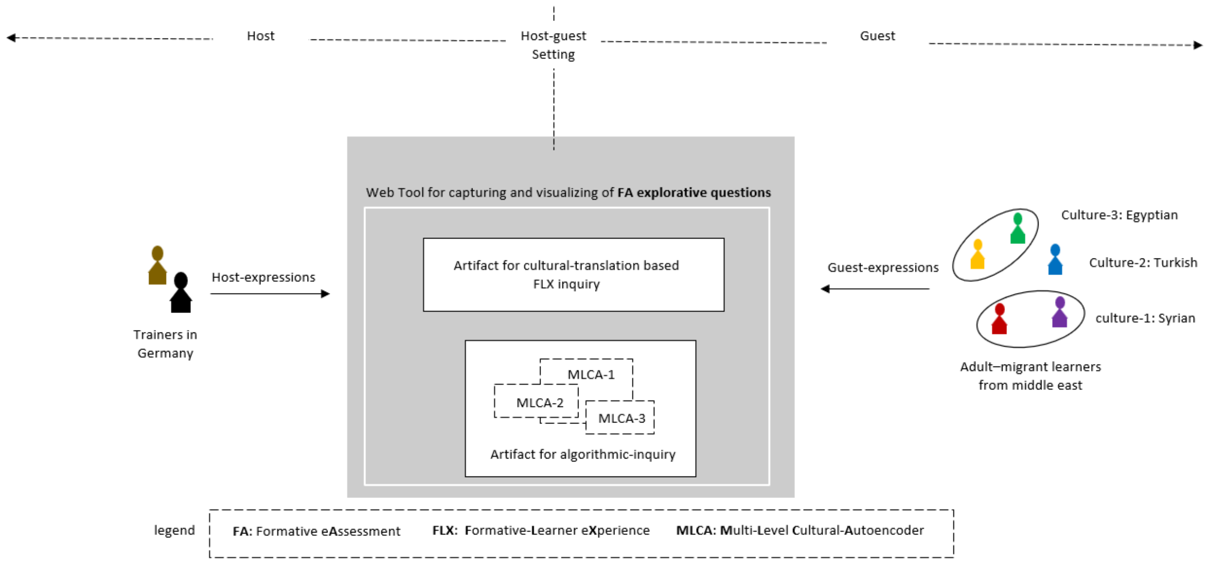

Fig. 5: operational-component

To facilitate preparing and planning for the test cases, Fig. 5 shows a target group of five adult-migrant learners from middle east. It shows also their national cultures (Egyptian, Turkish and Syrian) as three guest-cultures that should be put under analysis for understanding the gust-cultural expressions in oppressive or repressive situations experienced by an adult-migrant learner. It shows also how the cultural guest-expressions of the adult-migrant learners and the cultural host-expressions of the German trainers are planned to be captured by the tool. The captured host and guest expressions are expressions of explorative questions that should be fed by the tool to the cultural translation based *FLX* inquiry. This *FLX* inquiry uses *MLCA* algorithmic-inquiry to enable the exploration of feasible cultural translations. The *MLCA* algorithmic-inquiry operates on researching the suitable *MLCA* algorithms.

Finally, initializing the two artifacts for cultural-translation based *FLX* inquiry and *MLCA* algorithmic-inquiry, clarified in Fig. 5, is the first step in preparing and planning the test cases. Planning the test cases is summarized in next section.

6 PLANNING THE TEST CASES

To plan a test case, the object-of-assessment "*UD* barriers" and its sub-objects "oppressions" and "repressions" should be prepared via initializing the required lab-inquiries. As an example for test case preparation, the preparation is presented and clarified in table 1.

Table 1. example of a test case preparation.

Object-of-assessment	***sub-objects of assessment***	***Required operational services***	***Required lab-inquiries***
UD barriers that face the adult-migrants group (see Fig. 5)	"Oppressions" that can be captured as stream of **unconscious** cultural-expressions	Capturing and visualizing the *FA* explorative questions at collective-level (see Fig. 5)	cultural-translation based *FLX* inquiry, and *MLCA* algorithmic-inquiry (see Fig. 5) (see preparation guidelines in sections 3, 4 and 5)
	"repressions" that can be captured as stream of **unconscious** cultural-expressions	Capturing and visualizing *FA* explorative questions at individual-level (see Fig. 5)	

To sum up, table 1 summarizes the initial steps for capturing and visualizing the *FA* explorative questions that are required to enable the advanced steps for capturing the streams of **unconscious** cultural-expressions as sub-objects of assessment. The sub-objects are "oppressions" and "regressions" that should be presented as single "UD barriers" object of assessment required to operate the lab-inquires for lab-inquiry based *FA* practicing. In scope of this work-on-progress paper, the lab-inquires are inquires for cultural-translation based *FLX* inquiry and *MLCA* algorithmic-inquiry. To prepare these inquiries, the preparation guidelines are presented and discussed in sections 3, 4 and 5. In next section, the conclusion and future works are presented.

7 CONCLUSION AND FUTURE WORKS

A concept of *UD* assessment suggested to serve *FA* designers in our inquiry-based workshops-setting is presented and discussed. The concept suggests designing the "*UD* barriers" as object-of-assessment to approach the unconscious deformations that hinder the adult-migrant-learners. The paper showed how the concept's theoretical and operational components are conceptualized as services to help *FA* designers approach the *UD* barriers and the issues of parallel low-literacies that hinder the adult-migrant-learners from building their self-assessment and self-formation. The paper showed how the theoretical-component (section 5.1) is grounded on integrating Miller's theoretical constructs and Paul Freire's theoretical constructs. It also showed how the operational-component (section 5.2) is designed as online web tool to enable capturing and visualizing *FA* explorative-questions to serve the lab-inquiry based *FA* practices. To highlight its scope as work-on-progress paper, the paper showed also how to prepare a test case for a lab-inquiry and its sub-inquires. The suggested test case has two sub-inquiries; namely a cultural-translation based *FLX* inquiry and *MLCA* algorithmic-inquiry (section 6). Preparing the third *SMSA* sub-inquiry is planned as first step in our future works. Our future works include also reporting how the suggested *UD* assessment systems can provide diagnostic functions that outperform the diagnostic functions in bloom's taxonomy based *FA* systems, like the systems in [17]. Moreover, the future works include enhancing the theoretical component by theoretical analytic constructs to enable analyzing streams of "projections" as streams of *UD* barriers. The streams of "projections" can be cultural-expressions captured in pre-migration or migration context. The "projections" cultural-expressions can be understood as "repressed content" collect in a therapy session designed based Jungian "shadow-work" psychology. Finally, to improve in our *FA* redesign workshops, the future works include revisiting the literacy curriculums and books for deep-learning and machine-learning algorithms like in [18, 19, 20] to serve researching how to design innovative algorithms to serve the adult-migrant cultural context.

Acronyms

SD: **S**oftware **D**evelopment	**MLCA**: **M**ulti-**L**evel **C**ultural-**A**utoencoder
FA: Formative e-**A**ssessment	**FLX**: Formative Learner eXperience
UX: **U**ser e**X**perience	**SMSA:** Software Modelling based Self-Assessment
UD: **U**nconscious-**D**eformation	**TLC: T**herapy, STEM **L**iteracy, **C**omposition

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